

Exam 2 Review

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Question 1

0/1 pt 3 19

For a standard normal distribution, find:

$$P(z < -0.73)$$

Express the probability as a decimal rounded to 4 decimal places.

Normal Distribution Widget

Question 2

0/1 pt 3 19

For a standard normal distribution, find:

$$P(z > -1.25)$$

Express the probability as a decimal rounded to 4 decimal places.

Distribution Widgets

Question 3

0/1 pt 3 19

For a standard normal distribution, find:

$$P(z < c) = 0.775$$

Find c rounded to two decimal places.

Question 4

0/1 pt 3 19

A normal distribution has a mean of 101 and a standard deviation of 4. Find the z-score for a data value of 113.

Use the formula $Z = \frac{x - \mu}{\sigma}$ where μ is the mean, σ is the standard deviation, and x is the data value.

_____ Round to 4 decimal places

● Question 5

0/1 pt 3 19

On a nationwide math test, the mean was 52 and the standard deviation was 4. If Roberto scored 75, what was his z-score?

Use the formula $Z = \frac{x - \mu}{\sigma}$ where μ is the mean, σ is the standard deviation, and x is Roberto's score. Round to 4 decimal places.

● Question 6

0/1 pt 3 19

About _____ square units of area is under the curve of the standard normal distribution but outside the interval $z = [-1, 1]$ (that is, beyond 1 standard deviations of the mean).

● Question 7

0/1 pt 3 19

Find the 55 -th percentile for a normal random variable $X \sim N(\mu = 48.7, \sigma = 6.3)$.

The 55-th percentile for X is _____ (Round the answer to 2 decimal places)

● Question 8

0/1 pt 3 19

According to a study, the time between taking a pain reliever and getting relief is normally distributed with a mean of 35 minutes and a standard deviation of 8.3 minutes. Let X be the time between taking a pain reliever and getting relief.

1. Describe the probability distribution of X and state its parameters μ and σ : X is

Select an answer ($\mu =$ _____, $\sigma =$ _____)

2. Find the probability that the time between taking a pain reliever and getting relief for a randomly selected patient is
- less than 49 minutes. _____ (Round the answer to 4 decimal places)
 - between 26 and 57 minutes. _____ (Round the answer to 4 decimal places)
 - more than 54 minutes. _____ (Round the answer to 4 decimal places)

Question 9

0/1 pt 3 19

In the US, the systolic blood pressure is normally distributed with a mean of 120 mmHg and a standard deviation of 25.3 mmHg. Let X be the systolic blood pressure of a randomly selected patient.

1. Describe the probability distribution of X and state its parameters μ and σ : X is Select an answer ($\mu =$ _____, $\sigma =$ _____)
2. Find the probability that the systolic blood pressure of a randomly selected patient is
- less than 130 mmHg. _____ (Round the answer to 4 decimal places)
 - between 92 and 118 mmHg. _____ (Round the answer to 4 decimal places)
 - more than 160 mmHg. _____ (Round the answer to 4 decimal places)

Question 10

0/1 pt 3 19

In the US, the systolic blood pressure is normally distributed with a mean of 125 mmHg and a standard deviation of 19 mmHg. Let 'X' be the systolic blood pressure of a randomly selected patient.

1. Describe the probability distribution of 'X' and state its parameters ' μ ' and ' σ ': 'X' is (' μ '= '_____', ' σ '= '_____')
2. Find the probability that the systolic blood pressure of a randomly selected patient is
 - a. less than 71 mmHg. _____ (Round the answer to 4 decimal places)
 - b. between 98 and 163 mmHg. _____ (Round the answer to 4 decimal places)
 - c. more than 111 mmHg. _____ (Round the answer to 4 decimal places)
3. Find the 25-th percentile for the systolic blood pressure. _____ mmHg (Round the answer to 1 decimal place)

Question 11

0/1 pt 3 19

Part 1 of 5

A poll was conducted by a researcher on March 18-19, 2020, in English and Spanish, among a random national sample of 249 adults. In the poll, 65 Americans approve of the president's management of the crisis. Construct a 90% confidence interval for the percentage of the Americans that approve of the president's management of the crisis.

- i. Procedure:
- ii. Assumptions: (select everything that applies)
 - ☐ The number of positive and negative responses are both greater than 10
 - ☐ Population standard deviation is known
 - ☐ Sample size is greater than 30
 - ☐ Simple random sample
 - ☐ Normal population
 - ☐ Population standard deviation is unknown
- iii. Unknown parameter:

Question 12

0/1 pt 3 19

You are a researcher studying the lifespan of a certain species of bacteria. A preliminary sample of 25 bacteria reveals a sample mean of $\bar{x}=72$ hours with a standard deviation of $s=6.4$ hours. You would like to estimate the mean lifespan for this species of bacteria to within a margin of error of 0.7 hours at a 98% level of confidence.

What sample size should you gather to achieve a 0.7 hour margin of error? Round your answer *up* to the nearest whole number.

n = _____ bacteria

● Question 13

0/1 pt 3 19

Part 1 of 5

An incubation period is a time between when you contract a virus and when your symptoms start. By surveying randomly selected local hospitals, a researcher was able to obtain a sample of incubation periods of 47 patients, and the sample average was found to be 5.83 days. Assume that the population standard deviation is 1.7 days. Construct a 90% confidence interval for the average incubation period of the novel coronavirus.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Population standard deviation is known
- ☐ Sample size is greater than 30
- ☐ Population standard deviation is unknown
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Simple random sample
- ☐ Normal population

iii. Unknown parameter:

● Question 14

0/1 pt 3 19

You have been studying your favorite candy, Hot Tamales. You measure the mass of 17 randomly chosen pieces. The data (units are grams) is displayed in the table below, and a histogram is reasonably symmetric. Estimate the mean mass of all Hot Tamales. Use a 90% confidence level.

3	3.14	2.85	3.37	3.03	3.26	2.88	3.02	2.79	2.88
2.77	2.61	2.48	2.82	2.59	3.07	2.74			

(Checksum: 49.3)

a) Establish that the normal distribution applies as a foundation.

b) Find the critical value and standard error.

Critical Value: _____ Standard Error: _____

c) Find the confidence interval: (_____, _____)

d) Interpret your 90% confidence interval in context.

Question 15

0/1 pt 3 19

A common claim is that garlic lowers cholesterol levels. In a test of the effectiveness of garlic, 19 subjects were treated with doses of raw garlic, and their cholesterol levels were measured before and after the treatment. The changes in their levels of LDL cholesterol (in mg/mL) have a mean of 1.48 mg/mL and a standard deviation of 19.49 mg/mL. Construct a 99% confidence interval for the average net change in LDL cholesterol after the garlic treatment. You may assume that the sample is from a normal population.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Sample size is greater than 30
- ☐ Population standard deviation is known
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Simple random sample
- ☐ Normal population
- ☐ Population standard deviation is unknown

iii. Unknown parameter:

● Question 16

0/1 pt 3 19

The proper operation of typical home appliances requires voltage levels that do not vary much. Listed below are 27 voltage levels (in volts) at a random house on 27 different days.

120.1	120.2	119.9	119.9	120.1
119.9	119.7	119.9	119.8	120.1
119.9	119.8	120.1	120	120.1
119.9	120	120.2	119.6	119.7
120	119.9	119.9	120.1	119.8
120	119.9			

The mean of the data set is 119.94; the sample standard deviation is 0.153; if the normality plot is not provided you may assume that the voltages are normally distributed.

Construct a 90% confidence interval for the variance (standard deviation) of all voltages in the house.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Population standard deviation is known
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Sample size is greater than 30
- ☐ Simple random sample
- ☐ Normal population
- ☐ Population standard deviation is unknown

iii. Unknown parameter:

● Question 17

0/1 pt 3 19

You are conducting a study to see if the accuracy rate for fingerprint identification is significantly more than 0.56. You use a significance level of $\alpha = 0.02$.

$$H_0 : p = 0.56$$

$$H_1 : p > 0.56$$

You obtain a sample of size $n=239$ in which there are 155 successes.

What is the test statistic for this sample? (Report answer accurate to three decimal places.)

test statistic = _____

What is the p-value for this sample? (Report answer accurate to four decimal places.)

p-value = _____

The p-value is...

- ☐ less than (or equal to) α
- ☐ greater than α

This test statistic leads to a decision to...

- ☐ reject the null
- ☐ accept the null
- ☐ fail to reject the null

As such, the final conclusion is that...

- ☐ There is sufficient evidence to warrant rejection of the claim that the accuracy rate for fingerprint identification is more than 0.56.
- ☐ There is not sufficient evidence to warrant rejection of the claim that the accuracy rate for fingerprint identification is more than 0.56.
- ☐ The sample data support the claim that the accuracy rate for fingerprint identification is more than 0.56.
- ☐ There is not sufficient sample evidence to support the claim that the accuracy rate for fingerprint identification is more than 0.56.

Question 18

0/1 pt 3 19

Part 1 of 5

In late March 2020, a research firm conducted a national poll of 1062 U.S. adults that asked their views on the federal COVID-19 economic stimulus package. The question was, "As you may know, Congress is considering a new economic stimulus package of at least 2 trillion dollars. Do you favor or oppose Congress passing this legislation?" Of those sampled, 578 favored passage. At the 1% significance level, do the data provide sufficient evidence to conclude that the percentage of U.S. adults that favor the passage is greater than 50%?

Procedure:

Assumptions: (select everything that applies)

- ☐ Population standard deviation is known
- ☐ Normal population
- ☐ Population standard deviation is unknown
- ☐ Sample size is greater than 30
- ☐ Simple random sample
- ☐ The number of positive and negative responses are both greater than 10

Question 19

0/1 pt 3 19

A college administrator claims that 80% of college students purchase their books from the campus bookstore. You think this is inaccurate and form a random sample of 128 students at that college and find that 91 of them purchased their books from the bookstore. Test the administrator's claim using a level of significance of 1%.

a. What type of test will be used in this problem?

Select an answer

b. What evidence justifies the use of this test? Check all that apply

- ☐ There are two different samples being compared
- ☐ The original population is approximately normal
- ☐ The population standard deviation is not known
- ☐ The population standard deviation is known
- ☐ The sample standard deviation is not known
- ☐ $np > 5$ and $nq > 5$
- ☐ The sample size is larger than 30

c. Enter the null hypothesis for this test.

H_0 :

d. Enter the alternative hypothesis for this test.

H_1 :

e. Is the original claim located in the null or alternative hypothesis?

Select an answer

f. What is the test statistic for the given statistics?

g. What is the p -value for this test?

h. What is the decision based on the given statistics?

Select an answer

i. What is the correct interpretation of this decision?

Using a _____ % level of significance, there sufficient evidence to the claim that 80% of college students purchase their books from the campus bookstore.

● Question 20

0/1 pt 3 19

The US Department of Energy reported that more than 61% of homes were heated by natural gas. A random sample of 375 homes in Oregon found that 254 were heated by natural gas. Test the claim using a 10% significance level.

a. What type of test will be used in this problem?

Select an answer

b. What evidence justifies the use of this test? Check all that apply

- ☐ The sample size is larger than 30
- ☐ The population standard deviation is not known
- ☐ The original population is approximately normal
- ☐ There are two different samples being compared
- ☐ The sample standard deviation is not known
- ☐ The population standard deviation is known
- ☐ $np > 5$ and $nq > 5$

c. Enter the null hypothesis for this test.

H_0 :

d. Enter the alternative hypothesis for this test.

H_1 :

e. Is the original claim located in the null or alternative hypothesis?

Select an answer

f. What is the test statistic for the given statistics?

g. What is the p -value for this test?

h. What is the decision based on the given statistics?

Select an answer

i. What is the correct interpretation of this decision?

Using a _____ % level of significance, there sufficient

evidence to Select an answer ▼ the claim that more than 61% of homes were heated by natural gas.

● Question 21

✔ 0/1 pt ↻ 3 ⇌ 19

Company records indicate that less than 18% of employees commute to work using public transportation. You suspect this is incorrect and for a sample of 187 employees, and find that 11% of them take public transportation to work. Test the claim using a 1% level of significance.

a. What type of test will be used in this problem?

Select an answer

b. What evidence justifies the use of this test? Check all that apply

- ☐ The sample size is larger than 30
- ☐ The population standard deviation is known
- ☐ The population standard deviation is not known
- ☐ The original population is approximately normal
- ☐ There are two different samples being compared
- ☐ $np > 5$ and $nq > 5$
- ☐ The sample standard deviation is not known

c. Enter the null hypothesis for this test.

$H_0:$ _____

d. Enter the alternative hypothesis for this test.

$H_1:$ _____

e. Is the original claim located in the null or alternative hypothesis?

Select an answer

f. What is the test statistic for the given statistics?

g. What is the p -value for this test?

h. What is the decision based on the given statistics?

Select an answer

i. What is the correct interpretation of this decision?

Using a _____ % level of significance, there Select an answer ▾ sufficient evidence to Select an answer ▾ the claim that less than 18% of employees commute to work using public transportation.

Question 22

0/1 pt 3 19

Part 1 of 5

A study assessed fatigue in Chicago public school teachers due to stress. A random sample of 31 teachers had the following post-work heart rates:

74.9	72.2	77.2	77.7	78.9
75.9	81.3	67.7	69.3	74.4
73.1	70.4	77.1	79.7	74.1
84.4	77.2	74.9	80.6	80.4
76.9	72.6	79	75.6	68.7
76.3	83.9	68.9	80.8	77
75				

Assume that the population standard deviation of post-work heart rates for Chicago public school teachers is 4.1 bpm. At the 10% significance level, do the data provide sufficient evidence to conclude that the average post-work heart rate for Chicago public school teachers is different from the normal resting heart rate of 71 bpm? (Note: The average and the standard deviation of the data are respectively 76 bpm and 4.31 bpm.)

Procedure: Select an answer ▾

Assumptions: (select everything that applies)

- ☐ Population standard deviation is unknown
- ☐ Population standard deviation is known
- ☐ Simple random sample
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Sample size is greater than 30
- ☐ Normal population

Question 23

0/1 pt 3 19

The Federal Bureau of Investigation (FBI) compiles information on robbery and property crimes by type and selected characteristics and publishes its findings in Population-at-Risk Rates and Selected Crime Indicators. According to that document, the mean value lost to purse snatching was 397 dollars in 2004. This year, the mean and standard deviation of the 16 randomly selected purse snatching offenses are 428.41 dollars and 34.38 dollars, respectively. At the 1% significance level, do the data provide sufficient evidence to conclude that this year's mean value lost to purse snatching is different from the mean value lost in 2004? You may also assume that the sample is from a normal population.

Procedure:

Assumptions: (select everything that applies)

- ☐ Simple random sample
- ☐ Population standard deviation is known
- ☐ Sample size is greater than 30
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Normal population
- ☐ Population standard deviation is unknown

● Question 24

0/1 pt 3 19

A sample of birth weights of 44 girls was taken. Below are the results (in g):

3037.4	3477.2	2642.7	3102.2	3711.7
3862.7	3310.4	3518.6	3805	3685.4
2877.4	4067.1	4029.3	3007.2	3489.2
2009	3275.2	4013.9	3009.2	4103.7
4091.7	3931.4	2894.6	3786.8	4204.4
2798.7	2418.9	3567.2	3316.3	3433.1
2966.1	3403.6	3170.9	2794.6	3305.7
4536.8	3051.2	3681.7	4017	3228.2
3660.2	3302.9	3158.2	2806.5	

(Note: The average and the standard deviation of the data are respectively 3399.1 g and 524.09 g.)

Use a 5% significance level to test the claim that the standard deviation of birthweights of girls is less than the standard deviation of birthweights of boys, which is 540 g.

Procedure:

Assumptions: (select everything that applies)

- ☐ Population standard deviation is unknown
- ☐ Sample size is greater than 30
- ☐ Population standard deviation is known
- ☐ Normal population
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Simple random sample

Question 25

0/1 pt 3 19

The numbers of false fire alarms were counted each month at a number of sites. The results are given in the following table.

Month	Number of Alarms
January	39
February	33
March	44
April	36
May	25
June	25
July	37
August	29
September	28
October	38
November	35
December	36

Test the hypothesis that false alarms are equally likely to occur in any month. Use 1% level of significance.

Procedure:

Assumptions: (select everything that applies)

- ☐ Simple random sample
- ☐ At most 20% of the expected frequencies are less than 5
- ☐ All expected frequencies are 1 or greater
- ☐ Independent samples
- ☐ Normal populations
- ☐ Equal population standard deviations

● Question 26

✓ 0/1 pt ↻ 3 ↺ 19

The table below shows the distribution of births across the days of the week in a random sample of 149 births from local records in a large city:

Days	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Births	21	26	18	16	24	26	18

Are births evenly distributed across the days of the week? Use 5% level of significance.

Procedure:

Assumptions: (select everything that applies)

- ☐ Normal populations
- ☐ Independent samples
- ☐ At most 20% of the expected frequencies are less than 5
- ☐ Equal population standard deviations
- ☐ All expected frequencies are 1 or greater
- ☐ Simple random sample

Step 1. Hypotheses Set-Up:

H_0 :	Select an answer
H_a :	Select an answer , and the test is Select an answer

Step 2. The significance level $\alpha =$ _____ %

Step 3. Compute the value of the test statistic using the table below: (Round the answers to 4 decimal places)

Category	[O]bserved	[E]xpected	$O - E$	$(O - E)^2$	$(O - E)^2 / E$
Sun	21		-0.2857	0.0816	0.0038
Mon	26	21.2857		22.2246	1.0441
Tue	18	21.2857	-3.2857		0.5072
Wed	16	21.2857	-5.2857	27.9386	
Thu	24	21.2857	2.7143	7.3674	0.3461
Fri	26	21.2857	4.7143	22.2246	1.0441
Sat	18	21.2857	-3.2857	10.7958	0.5072

Total:			Select an answer ▼	$\chi^2 = \sum (O - E)^2 / E =$	
				df =	

Step 4. Testing Procedure: (Round the answers to 3 decimal places)

CVA	PVA
Provide the critical value(s) for the Rejection Region:	Compute the P-value of the test statistic:
left CV is and right CV is	P-value is

Step 5. Decision:

CVA	PVA
Is the test statistic in the rejection region?	Is the P-value less than the significance level?

Conclusion:

Step 6. Interpretation:

At 5% significance level we have sufficient evidence to reject the null hypothesis in favor of the alternative hypothesis.

Question 27

0/1 pt 3 19

You are conducting a test of homogeneity for the claim that two different populations have the same proportions of the following two characteristics. Here is the sample data.

Category	Population #1	Population #2
A	18	54
B	41	13

The expected observations for this table would be

Category	Population #1	Population #2
A		
B		

The resulting squared Pearson residuals are:

Category	Population #1	Population #2
A		
B		

What is the chi-square test-statistic for this data?

$\chi^2 =$

Report all answers accurate to three decimal places. (Though you may need to retain more decimal places in the residuals to get the correct chi-square test statistic.)

Hint 1